



FPT Software Human Resource Management System - Recruiter Process Concentration

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Executive summary

FPT Software is an FPT corporation's subsidiary and a leading global IT services provider headquartered in Hanoi, Vietnam. Established in 1999, the company has expanded its presence across 27 countries with offices in major markets, including the USA, Japan, Germany, and Singapore. Known for delivering software solutions and services across diverse industries, FPT Software has grown rapidly, leading to an increase in the complexity of managing its workforce (Leading in Digital Transformation & IT Consulting Services, 2025).

The recruitment process at FPT Software is essential to the growing requirement to acquire top-tier IT expertise, but it's currently struggling with notable inefficiencies and obstacles. The current recruitment process is time-consuming, resulting in delayed onboarding and technically missed chances. Many tasks, like entering candidate information, scheduling interviews, combining evaluations, and communicating with applicants, are still done manually. This reliance on manual workflows often leads to delays, mistakes, and inconsistencies. As a result, operational efficiency drops, decision-making is affected, and candidates don't have a smooth experience. It is obvious that a slothful recruitment process and poor employee experience steadily affect FPT software's ability to attract and retain top-talent resources, directly or indirectly impacting their core value of innovation and competitiveness (Leading in Digital Transformation & IT Consulting Services, 2025).

The urge for an HRM solution is undeniable. To tackle these challenges, we suggest implementing a centralized Human Resource Management System (HRMS) designed to meet FPT recruitment processes. The process includes some tasks such as candidate tracking, interview scheduling, offer management, and onboarding processes (Leading in Digital Transformation & IT Consulting Services, 2025).

To give a thorough comprehension of the process, we analyze the detailed recruitment process, visualized through a Business Process Model and Notation (BPMN) diagram. The system design features a well-defined Entity-Relationship (ER) diagram that lays out the database structure, guaranteeing a logical flow of data throughout the recruitment journey. We also confirm the system's feasibility by implementing a structured database using SQL. To enhance the database, a prototype user interface (UI) is created to ensure a smooth user experience.

Sophisticated human resource management systems developed with database implementation showcases the potential of enterprise information systems that can help businesses like FPT Software to not only tackle the current recruitment problems but also sets the stage for ongoing improvement and innovation in managing human resources.

System design

Swimlane – diagram:

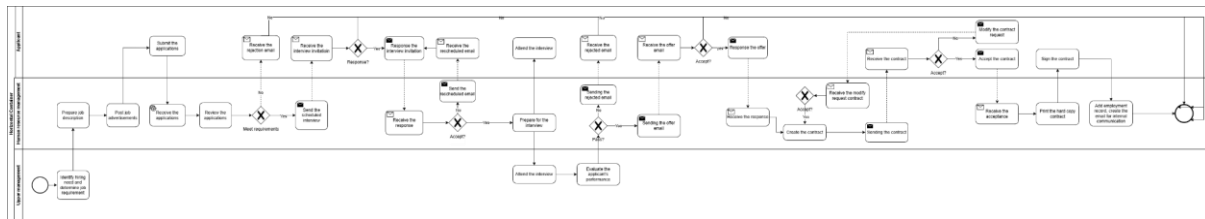


Figure 1 Swimlane - diagram for Business Process Modelling

The recruitment process in our project is designed to optimize the experience for three main user groups: **Applicants**, **Recruiters/HR**, and **Upper Management**. The BPMN model we developed illustrates in detail the steps of the process, from identifying recruitment needs to finalizing contracts with successful candidates.

Applicants can access the FPT Software website to search for suitable positions, register an account, or log in to submit their applications. Recruiters receive these applications and proceed to screen them based on predefined criteria. Qualified candidates are shortlisted for interviews, while those who do not meet the requirements are sent rejection emails.

During the interview phase, recruiters create interview schedules for the shortlisted candidates. These schedules are displayed on the interfaces of both applicants and upper management to ensure everyone is informed of the time and location. If the schedule changes, recruiters send notifications through the system. Both applicants and upper management attend the interviews as scheduled.

Once the interviews are completed, the system transitions to the evaluation phase. On the interfaces of recruiters and upper management, a list of candidates is displayed with the status marked as "not yet evaluated." Both groups collaborate to evaluate the candidates' performance based on mutually agreed criteria. The interface facilitates a smooth evaluation process with clear steps and functions tailored to each role.

Based on the evaluation results, recruiters decide to either accept or reject each candidate. For those who pass, recruiters send offer letters. When candidates accept the offer, the process moves to the contract phase. Recruiters create the employment contracts and upload them to the system. Applicants can view the contracts on their interfaces, along with a chat box icon to provide their feedback (accept, reject, or request modifications). If modifications are requested, the contracts are adjusted and sent back for final approval.

Once applicants approve the contracts, the signing process takes place. Applicants receive a hard copy of the signed contract, and recruiters update their profiles in the system to complete the process. Additionally, the system sends email notifications to internal departments to ensure that all relevant information is synchronized.

From our analysis of the BPMN model, we can infer that the previous recruitment process at FPT Software had several bottlenecks due to its heavy reliance on manual operations in most stages. Tasks such as application screening, managing interview schedules, sending invitations or rejection emails, evaluating candidates, and creating contracts were all done manually. This dependency on manual work increased the risk of errors and reduced overall efficiency.

Our BPMN model highlights these inefficiencies and emphasizes the need for a **centralized data system**. Such a system would automate many stages of the recruitment process, reduce the likelihood of errors, and improve processing speed. It would also provide a unified platform for managing all recruitment data, making it easier for recruiters, upper management, and applicants to track and complete the process seamlessly.

Entity – Relationship (ER) Diagram:

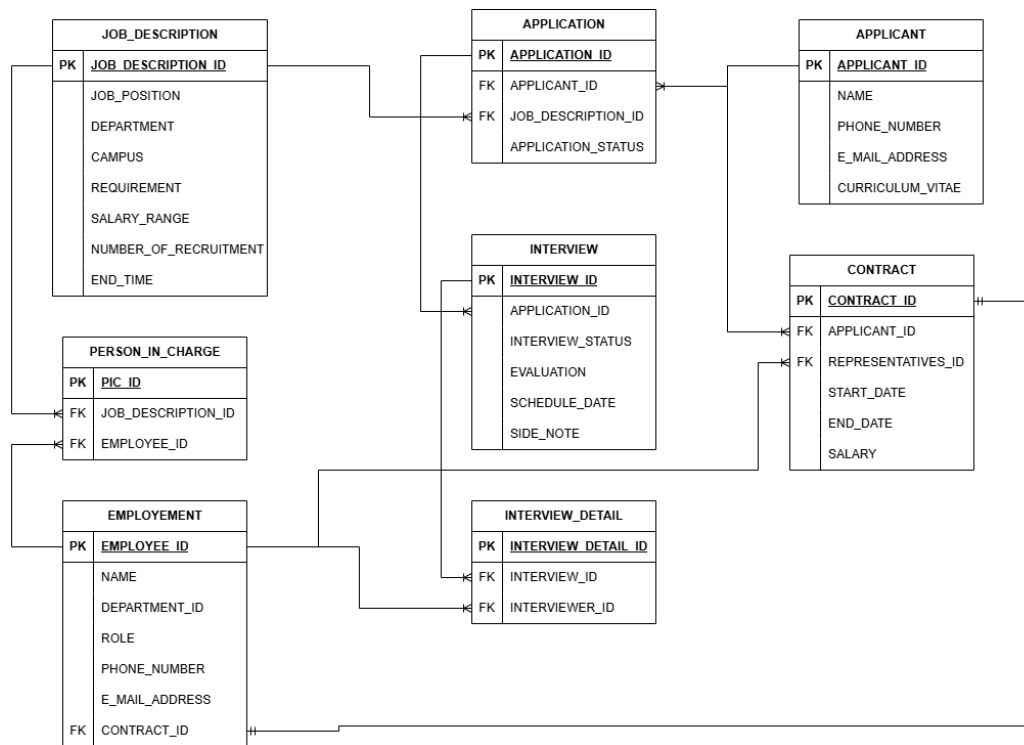


Figure 2 Entity Relationship Diagram

The ERD demonstrates the design of a database system for managing key HR operations, including recruitment, application processing, interviews, and employee management. Below is a breakdown of the entities and their relationships:

Entities and Attributes:

Strong Entities:

- **JOB_DESCRIPTION:** Contains details about job positions, including job title, department, campus, requirements, salary range, number of recruits, and job posting end date.
- **APPLICANT:** Holds applicant information such as name, contact details (phone number and email), gender, education, certifications, skills, and experience
- **EMPLOYEE:** Stores employee records, including name, role, department, contact details, and the associated contract.
- **CONTRACT:** Represents employment contracts, containing data about contract terms, representatives, start and end dates, and salaries.
- **APPLICATION:** Tracks job applications, connecting applicants to job descriptions and storing the current status of their applications.

Weak Entities:

- **INTERVIEW:** Depends on APPLICATION, manages the interview process, recording the status, evaluation, schedule, and additional notes for each interview.
- **INTERVIEW_DETAIL:** Depends on INTERVIEW, captures specific interviewer details for each interview session.
- **PERSON_IN_CHARGE:** Depends on JOB_DESCRIPTION AND EMPLOYEE, links specific employees to job postings, and designates responsibility for recruitment processes.

The strong entities will be linked using bridge tables:

- **PERSON_IN_CHARGE:** A bridge table between **EMPLOYEE** and **JOB_DESCRIPTION**, tracking employees responsible for managing specific jobs.
- **APPLICATION:** A bridge table between **APPLICANT** and **JOB_DESCRIPTION**, tracking which applicant applied for which job and the application status.
- **INTERVIEW_DETAIL:** A bridge table between **INTERVIEW** and the interviewers (**EMPLOYEE**), storing detailed information about each interviewer participating in an interview session.

The data flow begins with the creation of new job descriptions. When a job is created, its details are stored in the **JOB_DESCRIPTION** table. Simultaneously, the employee responsible for the job is recorded in the **PERSON_IN_CHARGE** table, establishing a link between **JOB_DESCRIPTION** and **EMPLOYEE**. In the next step, when an applicant submits a job application, their personal information is stored in the **APPLICANT** table if it is not already present. The application details are then recorded in the **APPLICATION** table, linking the applicant (**APPLICANT**) and the job (**JOB_DESCRIPTION**).

Once an application is approved, the interview process is initiated. The interview schedule and related details are stored in the **INTERVIEW** table. Additionally, specific information about each interviewer is recorded in the **INTERVIEW_DETAIL** table, which links to **EMPLOYEE**. Finally, after the interview succeeds, the contract details are created and stored in the **CONTRACT** table, linking back to the applicant in the **APPLICANT** table. Once the applicant officially becomes an employee, their information is added to the **EMPLOYEE** table, establishing a connection with the **CONTRACT** table. This process ensures seamless data flow and integration across all stages of recruitment and employment.

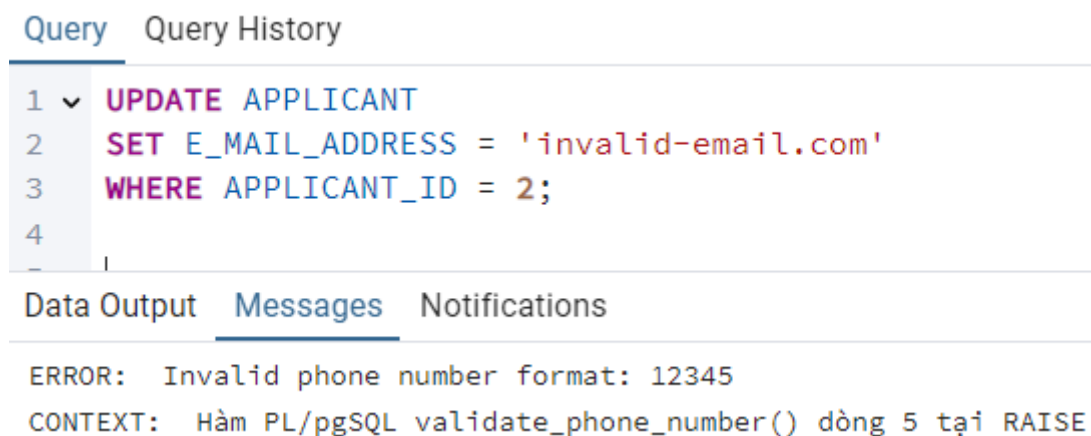
Proof of Concept (POC): Database implementation

SQL Queries for database establishment:

Based on the above ERD, we use the `CREATE TABLE IF NOT EXISTS` command to create the tables for storing user information. Each table is designed with appropriate fields and data types to store specific information, and relationships between tables are enforced using `FOREIGN KEY` constraints. These constraints ensure data integrity and allow cascading actions such as `ON DELETE CASCADE`, maintaining consistency by deleting dependent records when the parent record is deleted.

Additionally, we use stored procedures and triggers in the database to automate calculations, validate input data, and efficiently handle changes. Triggers are activated when users perform `INSERT`, `UPDATE`, or `DELETE` operations, ensuring the data remains valid, logical, and consistent with business processes.

Email Validation: Triggers such as `trigger_validate_applicant_email` and `trg_validate_employee_email` are used to check the email format and ensure there are no duplicates in the `APPLICANT` and `EMPLOYEE` tables.



The screenshot displays a database query editor interface. At the top, there are two tabs: 'Query' and 'Query History'. The 'Query' tab is active, showing an SQL statement with line numbers 1 through 4. The statement is: `1 UPDATE APPLICANT`, `2 SET E_MAIL_ADDRESS = 'invalid-email.com'`, `3 WHERE APPLICANT_ID = 2;`, and `4` followed by a cursor. Below the query editor, there are three tabs: 'Data Output', 'Messages', and 'Notifications'. The 'Messages' tab is active, showing an error message: `ERROR: Invalid phone number format: 12345` and a context message: `CONTEXT: Hàm PL/pgSQL validate_phone_number() dòng 5 tại RAISE`.

```
Query Query History
1 UPDATE APPLICANT
2 SET E_MAIL_ADDRESS = 'invalid-email.com'
3 WHERE APPLICANT_ID = 2;
4
Data Output Messages Notifications
ERROR: Invalid phone number format: 12345
CONTEXT: Hàm PL/pgSQL validate_phone_number() dòng 5 tại RAISE
```

Figure 3 Trigger email validation

Phone Number Validation: Triggers like `check_employee_phone_number` and `check_applicant_phone_number` validate phone numbers using regex to ensure correct format.

Query	Query History
1	▼ UPDATE APPLICANT
2	SET PHONE_NUMBER = '12345'
3	WHERE APPLICANT_ID = 1;
4	

Data Output	Messages	Notifications
ERROR: Invalid phone number format: 12345		
CONTEXT: Hàm PL/pgSQL validate_phone_number() dòng 5 tại RAISE		

Figure 4 Trigger phone number validation

Time Validation: Triggers like `validate_job_description_end_time` and `validate_interview_schedule_date` ensure that time values (e.g., `END_TIME` or `SCHEDULE_DATE`) are in the future.

1	▼ UPDATE JOB_DESCRIPTION
2	SET END_TIME = '2024-01-01 10:00:00'
3	WHERE JOB_DESCRIPTION_ID = 5;
4	

Data Output	Messages	Notifications
ERROR: END_TIME must be in the future: 2024-01-01 10:00:00		
CONTEXT: Hàm PL/pgSQL validate_job_description_timestamp() dòng 4 tại RAISE		

Figure 5 Trigger time validation

Automatic Calculations: The `update_evaluation_trigger` automatically calculates the average evaluation value (EVALUATION) based on partial evaluations (`partial_evaluation`) from the INTERVIEW_DETAIL tableA. The `check_partial_evaluation` trigger ensures the evaluation value is between 0 and 100.

	application_id integer	interviewer_id integer	name character varying (50)	partial_evaluation integer	evaluation integer
1	1	23	Alice Nguyen	46	38
2	1	33	Dory	30	38
3	2	23	Alice Nguyen	70	79
4	2	24	Bob Tran	87	79
5	3	24	Bob Tran	10	44
6	3	25	Charlie Le	78	44
7	4	25	Charlie Le	70	79
8	4	26	Diana Vu	88	79
9	5	26	Diana Vu	93	69

Figure 6 Interview detail table view

```

8
9  ✓ UPDATE INTERVIEW_DETAIL
10 SET PARTIAL_EVALUATION = 150
11 WHERE INTERVIEW_DETAIL_ID = 20 ;
12

```

Data Output Messages Notifications

```

ERROR:  Partial evaluation must be between 0 and 100: 150
CONTEXT:  Hàm PL/pgSQL validate_partial_evaluation() dòng 4 tại RAISE

```

Figure 7 Trigger evaluation range validation

Schedule Conflict Detection: Triggers like `check_interview_schedule` and `check_interviewer_schedule` check if there are scheduling conflicts for candidates or interviewers during interview scheduling.

Update Procedures: The `update_schedule_date` procedure allows automatic updating of interview times to simplify schedule management. Procedures like `update_application_status` and `update_interview_status` are used to change the status of applications or interviews, ensuring that data is updated accurately.

Testing and Validation:

Our system provides HR managers with comprehensive tools to streamline the recruitment process. It allows for easy tracking and analysis of recruitment data, from monitoring applicants' progress to checking historical data. Data integrity is ensured through automated validation checks, such as email and phone number formatting, and time validation

for job descriptions and interviews. The system also automates calculations, like averaging interview evaluations, and updates the status of applications and interviews to reduce manual input errors. It helps detect scheduling conflicts for interviews, ensuring smooth coordination. Additionally, HR managers can easily generate reports and make data-driven decisions based on detailed queries, optimizing the entire recruitment workflow.

Proof of Concept (POC): User Interface (UI)

Overview of the prototype:

The recruitment processing prototype is designed to streamline the hiring workflow for recruiters, upper management, and applicants through an intuitive and sequential user interface. It incorporates three distinct user roles: recruit, applicant, and upper management. Each role is supported by tailored screens and features that facilitate efficient collaboration and decision-making throughout the recruitment lifecycle (New employees, 2025).

The process begins with the recruiter's screen, from job posting and application submission to applicant evaluation and contract management. By providing logical function design based on recruitment processing, and structured workflows, this system improves experience for recruiters. Besides that, applicants can easily access an intuitive interface to submit applications, attend interviews, and track contract statuses. Upper management plays a key role in checking, attending interviews, and reviewing evaluations. This prototype not only simplifies the complex hiring process, but also promotes transparency, objectivity, and collaboration between individuals, ensuring that the hiring process is smooth, easy, and efficient for all parties involved.

Features demonstrated:

Screen	Features	Explanation
Recruit	Filter search, create new position, upload job position	Employers easily find suitable candidates and set up new job positions with the necessary documentation.
	Check the status of other recruiters	View other recruiters' tasks to get an overview of the task
	Check the status of the job: to recruit, date end,...	Check time is left to close a job, expired jobs will be automatically pushed down.

	Shortlist applicants based on qualifications (accept or reject)	Narrow down candidate lists based on specific criteria to increase screening efficiency.
	Search, and filter applicants	Search for candidate information and sort application information in order such as scores,...
	Check & set the schedule	Helps employers easily find suitable candidates and set up new job positions with the necessary documentation.
	View and evaluate applicants after interviews	Provide candidate information and support direct post-interview assessment for data storage.
	Decide on acceptance or rejection, and export evaluations	Allows storing and exporting assessment results to make decisions about interview approval
	Download, edit, and upload contracts	Support contract management between candidates and companies, including flexible editing and sending.
	Check status contract	Show contract status (submitted, rejected, etc.) to track progress.
Applicants	Sign In to the System	Provide access to the system to start the application process and track job progress.
	Alternative Login Methods	Quick login via Facebook or Google saves time and offers applicants convenience.
	Fill Out Personal Information	Applicants fill in personal details such as full name, phone number, email, gender, education level, certifications,

		skills, and work experience.
	Submit Application	Click the " Submit Now " button to send their application. Their information will be transferred to the recruitment system for processing and storage.
	View Applied Jobs	View other positions they've applied for, including job title, application date, assigned recruiter, and the current status (Initial Qualification, Interview, Contract, Fail).
	Receive Interview Schedules	Provide detailed information about interview times and locations.
	Respond to Interview Schedules	Applicants can choose " Attend " or " Reject " the interview schedule if the time doesn't fit their availability.
	View and Respond to Contracts	Applicants can choose to " Accept " or " Reject " the job contract. If they reject it, they can provide specific reasons for the recruiter to consider and possibly adjust the terms.
Upper Management	Check Interview Schedules	Managers can review and track interview schedules for applicants, including time details and applicant names.
	Respond to Interview Invitations	Managers can select " Attend " or " Reject " an interview if the schedule isn't suitable. In case of rejection, they can coordinate with recruiters to adjust the interview schedule.

	View and Evaluate Applicants After Interviews	Provide candidate information and support direct post-interview assessments for data storage.
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Conclusion and Future Orientation:

Summary of Achievements:

The proposed Human Resource Management System for FPT Software overcomes existing problems in the recruitment process by moving away from manual operations. By utilizing BPMN analysis, Entity-Relationship diagrams and database implementation, the system achieves significant achievements like smoother workflows, organized data management, and enhanced operational efficiency through automated features such as calculating interviewer scores. Moreover, the user-friendly prototype showcases features like applicant tracking, evaluation management, and reporting, proving that the system is both practical and effective in improving recruitment operations.

Next steps:

Moving forward with FPT Software's suggested Human Resource Management system, the next step is to develop the current prototype into a fully functional HRMS. This will feature a strong back-end and ensure real-time data synchronization while being seamlessly integrated with the company's existing IT setup. We'll conduct user testing involving HR staff and hiring managers to gather feedback on the system's features, design, and overall performance for further enhancements. We also plan to carry out advanced features like predictive analytics and machine learning models to help with applicant ratings, alongside implementing security measures to guarantee compliance with data protection regulations.

Vision for full-implementation:

The goal of deploying FPT Software's Human Resource Management system is to create a comprehensive recruiting ecosystem that manages every aspect of the hiring process, from job postings to onboarding. This solution will provide real-time insights and data-driven recommendations to support strategic workforce planning, enabling managers to make better-informed decisions. Additionally, it also provides the groundwork for larger HR digitization initiatives, establishing FPT Software as a market leader in tech-enabled people management.

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TEAM CONTRIBUTION STATEMENT

Tasks \ Team Members	Đào Đức Anh	Lưu Hà Linh	Phạm Thị Mỹ Dung	Lê Thị Hải Ân	Nguyễn Thị Thảo
Team and supervisor meetings	8	8	8	8	8
Client meetings (preparation, attending, follow up)	2	2	2	2	2
Team management and administration	6	6	6	6	6
Research searching, analysing, modelling	12	12	12	12	12
Software evaluation, development, testing	15	15	15	15	15
Presentations Preparing, delivering, revising	0.5	0.5	0.5	0.5	0.5
Team Documents Writing	2	2	2	2	2
Reviewing Checking and correcting other's work	3	3	3	3	3
Total hours for each time period	48.5	48.5	48.5	48.5	48.5
Percentage	20%	20%	20%	20%	20%

TEAM ID:3

- State the number of hours spent
- It is not necessary that a member contributes in every area. It is a member's total contribution to the project for the entire project that is important.
- Round off contributions to tasks to nearest hour
- Do not include time spent on individually assessed tasks such as Journals

- All members must agree to and sign
- Record the end date of the time period for which you are signing
- The supervisor will sign off and record the date on which he/she signed off when satisfied that the contributions appear to fairly represent each person's time contribution to team based tasks.

Member's Name	Members Signature
A: Dennis Nguyen	DucAnh
B:	HaLinh
C:	Dung
D:	HaiAn
E:	Thao
End date of Period	
Supervisors Signature	
Date	